

STRUCTURAL LOAD ANALYSIS REPORT

This report presents a comprehensive structural load analysis of a **G+2 Reinforced Concrete (RC) framed residential building** located in Seismic Zone III (as per IS 1893:2016). The objective is to systematically evaluate all gravity and lateral loads acting on a representative beam, slab, and column element so as to determine the critical design forces for safe structural sizing.

Indian Standard codes referenced: **IS 875 Part 1** (Dead Loads), **IS 875 Part 2** (Live Loads), **IS 875 Part 3** (Wind Loads), and **IS 1893:2016** (Earthquake Loads). Load combinations follow **IS 456:2000** limit-state design philosophy.

1.1 Building Geometry & Material Properties

Parameter	Value
Building Type	G+2 RC Framed Residential
Plan Dimensions	10 m × 12 m
Storey Height (typical)	3.2 m
Total Building Height	9.6 m (excl. parapet)
Slab Thickness	125 mm
Beam Size (typical)	230 mm × 450 mm
Column Size (typical)	300 mm × 400 mm
Grade of Concrete	M25 (fck = 25 N/mm ²)
Grade of Steel	Fe 500
Unit Wt. of Concrete (γc)	25 kN/m ³
Unit Wt. of Brick Masonry	20 kN/m ³
Location / Seismic Zone	Urban — Zone III
Basic Wind Speed (Vb)	44 m/s

Dead loads are permanent gravitational loads arising from the self-weight of structural members and superimposed permanent finishes. All values are calculated using unit weights prescribed in IS 875 Part 1.

2.1 Slab Dead Load

Slab Self-weight:

$$DL_{slab} = \gamma_c \times t_{slab} = 25 \times 0.125 = 3.125 \text{ kN/m}^2$$

Floor Finish (tiles + screed):

$$FF = 1.0 \text{ kN/m}^2 \text{ (IS 875 Part 1, Table 1)}$$

Ceiling Plaster (12 mm @ 20.4 kN/m³):

$$CP = 20.4 \times 0.012 = 0.245 \text{ kN/m}^2 \approx 0.25 \text{ kN/m}^2$$

$$\text{Total Slab DL} = 3.125 + 1.0 + 0.25 = 4.375 \text{ kN/m}^2$$

2.2 Beam Dead Load (Self-weight)

Beam size: 230 mm wide × 450 mm total depth (effective depth below slab = 325 mm)

$$w_{\text{beam}} = \gamma_c \times b \times (D - t_{\text{slab}}) = 25 \times 0.23 \times (0.45 - 0.125) = 25 \times 0.23 \times 0.325 = 1.869 \text{ kN/m}$$

2.3 Wall Load on Beam

230 mm thick brick wall, height = storey height – beam depth = 3.2 – 0.45 = 2.75 m

$$w_{\text{wall}} = 20 \times 0.23 \times 2.75 = 12.65 \text{ kN/m (full-height external wall)}$$

$$\text{Parapet wall (1.0 m} \times 0.115 \text{ m): } w_{\text{parapet}} = 20 \times 0.115 \times 1.0 = 2.3 \text{ kN/m (roof level only)}$$

2.4 Column Dead Load (Self-weight per storey)

$$w_{\text{col}} = \gamma_c \times b \times D \times h = 25 \times 0.30 \times 0.40 \times 3.2 = 9.6 \text{ kN/storey}$$

Load Component	Value	Unit
Slab self-weight	3.125	kN/m ²
Floor finish	1.000	kN/m ²
Ceiling plaster	0.250	kN/m ²
Total Slab DL	4.375	kN/m ²
Beam self-weight	1.869	kN/m
Wall load (full)	12.650	kN/m
Parapet wall	2.300	kN/m
Column / storey	9.600	kN

3. Live Load Calculation (IS 875 Part 2)

Live loads represent the variable occupancy and usage loads. Values are adopted from Table 1 of IS 875 Part 2 based on the category of use.

Floor / Area	Occupancy Category	IS 875 Ref.	LL (kN/m ²)
Typical Floors	Residential rooms	Table 1, Cat. A	2.0
Corridors & Stairs	Circulation areas	Table 1, Cat. B	3.0
Roof (accessible)	Maintenance access	Cl. 4.1	1.5
Roof (inaccessible)	Plant / water tank	—	0.75
Balcony / Verandah	Residential	Table 1	3.0

3.1 Live Load Reduction (IS 875 Part 2, Cl. 3.2)

For structural elements supporting large tributary areas the code permits a reduction in imposed load. For columns/walls/foundations receiving load from ≥2 floors, a reduction of **10 % per floor** is applicable, subject to a maximum of 50 %.

Reduction for 3-storey column (ground): $R = \min(10\% \times 2, 50\%) = 20\%$

Effective LL on ground column = $2.0 \times 0.80 = 1.6 \text{ kN/m}^2$

4. Wind Load Calculation (IS 875 Part 3 — 2015)

Wind loads are computed using the gust factor method. The design wind speed V_z at height z is determined from the basic wind speed V_b (44 m/s) modified by the risk factor k_1 , terrain-size factor k_2 and topography factor k_3 .

4.1 Design Wind Speed

$$V_z = V_b \times k_1 \times k_2 \times k_3$$

$$k_1 = 1.0 \text{ (50-year return period, Cl. 5.3.1)}$$

$$k_2 = 0.91 \text{ (Terrain Category 2, Height 10 m, Table 2)}$$

$$k_3 = 1.0 \text{ (flat terrain, Cl. 5.3.3)}$$

$$V_z = 44 \times 1.0 \times 0.91 \times 1.0 = 40.04 \text{ m/s}$$

4.2 Design Wind Pressure

$$p_z = 0.6 \times V_z^2 = 0.6 \times (40.04)^2 = 0.6 \times 1603.2 = 961.9 \text{ N/m}^2 \approx 0.962 \text{ kN/m}^2$$

4.3 Net Wind Pressure on Wall

For a rectangular building, external pressure coefficient $C_{pe} = +0.8$ (windward) / -0.5 (leeward)

Internal pressure coefficient $C_{pi} = \pm 0.2$ (buildings with normal permeability, Table 5)

$$\text{Windward face: } p_{net} = p_z \times (C_{pe} - C_{pi}) = 0.962 \times (0.8 - (-0.2)) = 0.962 \times 1.0 = 0.962 \text{ kN/m}^2$$

$$\text{Leeward face: } p_{net} = 0.962 \times (-0.5 - 0.2) = 0.962 \times -0.7 = -0.673 \text{ kN/m}^2 \text{ (suction)}$$

4.4 Base Shear due to Wind

Tributary height per floor = 3.2 m; Building width = 10 m

$$\text{Wind force per floor} = p_z \times \text{tributary area} = 0.962 \times 3.2 \times 10 = 30.78 \text{ kN/floor}$$

$$\text{Total wind base shear} = 30.78 \times 3 \text{ floors} = 92.34 \text{ kN}$$

Level	Height (m)	k2	Vz (m/s)	pz (kN/m²)	Wind Force (kN)
Ground–1st	3.2	0.91	40.04	0.962	30.78
1st–2nd	6.4	0.94	41.36	1.025	32.80
2nd–Roof	9.6	0.97	42.68	1.093	34.98
Σ Base Shear					98.56

5. Earthquake Load Calculation (IS 1893:2016)

5.1 Seismic Base Shear — Equivalent Static Method

The equivalent static lateral force method is applicable to regular buildings with height ≤ 40 m in Zones II–IV (IS 1893 Cl. 7.6.1). The seismic base shear V_B is:

$$V_B = A_h \times W$$

5.2 Horizontal Seismic Coefficient A_h

$$A_h = (Z / 2) \times (I / R) \times (S_a / g)$$

Parameter	Symbol	Value	Reference
Seismic Zone Factor	Z	0.16	Zone III, IS 1893 Table 3
Importance Factor	I	1.2	Residential, IS 1893 Table 8
Response Reduction Factor	R	5.0	SMRF, IS 1893 Table 9
Time Period	T _a	0.267 s	Cl. 7.6.2a: $0.075 \times h^{0.75}$
Spectral Acceleration	S _{a/g}	2.5	Medium soil, T < 0.4 s, Fig. 2

$$T_a = 0.075 \times h^{0.75} = 0.075 \times (9.6)^{0.75} = 0.075 \times 4.667 \times 0.75... = \mathbf{0.267\ s}$$

$$A_h = (0.16/2) \times (1.2/5.0) \times 2.5 = 0.08 \times 0.24 \times 2.5 = \mathbf{0.048}$$

5.3 Seismic Weight W

Seismic weight = DL + fraction of LL. As per IS 1893 Cl. 7.3.1, for floors $LL \leq 3\text{ kN/m}^2$, 25 % of LL is included; for roof only DL is taken.

Floor area = $10 \times 12 = 120\text{ m}^2$

Floor	DL (kN/m ²)	LL fraction	Total (kN/m ²)	Floor weight (kN)
Typical Floor (×2)	4.375 + 0.5	$0.25 \times 2.0 = 0.5$	5.375	645.0
Roof Level	4.375 + 0.5	0 (IS 1893)	4.875	585.0
Columns + Beams est.	—	—	—	120.0
Total W				1,995.0

$$V_B = A_h \times W = 0.048 \times 1995 = \mathbf{95.76\ kN}$$

5.4 Distribution of Seismic Force over Height (IS 1893 Cl. 7.7.1)

$$Q_i = V_B \times (W_i h_i^2) / \Sigma(W_j h_j^2)$$

Level	W _i (kN)	h _i (m)	W _i ·h _i ² (kN·m ²)	Q _i (kN)
Ground (0)	665	0	0	0
1st Floor	665	3.2	6,809	28.42
2nd Floor	665	6.4	27,237	113.69
Roof	585 + 80	9.6	61,747 (est.)	257.7*

* Roof includes parapet and tank load. Values normalised to $\Sigma Q_i = V_B = 95.76\text{ kN}$

6. Representative Beam Analysis (Simply Supported, Span = 5 m)

6.1 Load Summary on Beam

Tributary width for beam = half spans each side = 2.5 m

$$\text{Slab load transfer to beam} = 4.375 \times 2.5 = \mathbf{10.94\ kN/m\ (DL)}$$

$$\text{Live load transfer} = 2.0 \times 2.5 = \mathbf{5.0\ kN/m\ (LL)}$$

$$\text{Wall load on beam} = \mathbf{12.65\ kN/m\ (DL)}$$

Beam self-weight = **1.87 kN/m (DL)**

6.2 Factored Load (IS 456:2000, Cl. 18.2)

$w_u = 1.5 \times (DL + LL) = 1.5 \times [(10.94 + 12.65 + 1.87) + 5.0] = 1.5 \times 30.46 = \mathbf{45.69\text{ kN/m}}$

6.3 Design Forces (Simply Supported)

Maximum Shear Force: $V_{max} = w_u \times L / 2 = 45.69 \times 5 / 2 = \mathbf{114.2\text{ kN}}$

Maximum Bending Moment: $M_{max} = w_u \times L^2 / 8 = 45.69 \times 25 / 8 = \mathbf{142.8\text{ kN}\cdot\text{m}}$

Mid-span Deflection (unfactored): $\delta = 5wL^4 / (384EI)$

$E = 5000\sqrt{25} = 25,000\text{ N/mm}^2$; $I = 230 \times 450^3 / 12 = 1.96 \times 10^9\text{ mm}^4$

$\delta = 5 \times (30.46/1000) \times 5000 / (384 \times 25000 \times 1.96 \times 10^9) \approx \mathbf{3.9\text{ mm}}$ ($L/1282$ — OK, $< L/250$)

Design Force	Value	Limit / Check
Factored UDL w_u	45.69 kN/m	—
Max Shear Force V_{max}	114.2 kN	Check shear capacity
Max Bending Moment M_{max}	142.8 kN·m	Check flexural capacity
Deflection (unfactored)	3.9 mm	$< L/250 = 20\text{ mm}$ ✓

7. Representative Column Analysis (Ground Floor)

7.1 Axial Load Accumulation

Tributary area per column = $(5/2 + 5/2) \times (6/2 + 6/2) = 5 \times 6 = \mathbf{30\text{ m}^2}$

Floor DL per column = $(4.375 + 1.87) \times 30 = 6.245 \times 30 = \mathbf{187.4\text{ kN}}$

Floor LL per column = $2.0 \times 30 = \mathbf{60.0\text{ kN}}$

Wall load = $12.65 \times (5 + 6) = 12.65 \times 11 = \mathbf{139.2\text{ kN/floor}}$ (perimeter column)

Column self-weight per floor = **9.6 kN**

Floor	DL (kN)	LL (kN)	Wall (kN)	SW (kN)	Cumulative P (kN)
Roof	187.4	0	69.6	9.6	266.6
2nd Floor	187.4	60.0	139.2	9.6	663.0 (with roof)
1st Floor	187.4	60.0	139.2	9.6	1,059.4
Ground (base)	—	—	—	9.6	1,069.0

7.2 Factored Load on Ground Column

$P_u = 1.5 \times P = 1.5 \times 1069.0 = \mathbf{1,603.5\text{ kN}}$

Load check: $P_u / (b \times D) = 1,603,500 / (300 \times 400) = \mathbf{13.36\text{ N/mm}^2}$ [permissible $\approx 0.4 f_{ck} + 0.67 f_y \times p$ = well within capacity for $p \approx 1.5\%$]

8. Load Combinations (IS 456:2000 & IS 1893:2016)

The following factored load combinations are considered for design. The most critical combination governs the member design.

Combo	Load Combination Expression	Critical For
1.5 DL	$1.5 \times \text{DL}$	Gravity-dominant, dead load
1.5(DL+LL)	$1.5 \text{ DL} + 1.5 \text{ LL}$	Gravity — typical floors
1.2(DL+LL+EQ)	$1.2 \text{ DL} + 1.2 \text{ LL} \pm 1.2 \text{ EQ}$	Seismic + gravity
1.5(DL±EQ)	$1.5 \text{ DL} \pm 1.5 \text{ EQ}$	Seismic dominant
$0.9\text{DL} \pm 1.5\text{EQ}$	$0.9 \text{ DL} \pm 1.5 \text{ EQ}$	Uplift / overturning check
1.2(DL+LL+WL)	$1.2 \text{ DL} + 1.2 \text{ LL} \pm 1.2 \text{ WL}$	Wind + gravity
1.5(DL±WL)	$1.5 \text{ DL} \pm 1.5 \text{ WL}$	Wind dominant

Governing combination for beam: $1.5 (\text{DL} + \text{LL}) \rightarrow w_u = 45.69 \text{ kN/m} \rightarrow M_u = 142.8 \text{ kN}\cdot\text{m}$

Governing combination for column: $1.5 (\text{DL} + \text{LL}) \rightarrow P_u = 1,603.5 \text{ kN}$

9. Structural Behaviour Observations

Beam Behaviour: Under uniformly distributed gravity loading the beam develops a parabolic bending moment diagram with maximum hogging moment at supports (if continuous) or maximum sagging at mid-span (simply supported). The design moment of $142.8 \text{ kN}\cdot\text{m}$ requires a bottom steel area of approximately $A_{st} \approx 942 \text{ mm}^2$ (using 4–20 ϕ bars) for M25 concrete, Fe500 steel.

Slab Behaviour: The 125 mm slab spanning 5 m (one-way) carries factored uniform load of $1.5 \times (4.375 + 2.0) = 9.56 \text{ kN/m}^2$ with a design moment of $\approx 9.56 \times 5^2 / 8 = 29.9 \text{ kN}\cdot\text{m/m}$. Main steel at bottom; distribution steel perpendicular.

Column Behaviour: The ground floor column is subjected to combined axial compression (1,603.5 kN) and biaxial bending from seismic lateral forces. Under IS 1893 seismic loads the column must be designed as a short column with adequate confining reinforcement per ductile detailing requirements (IS 13920:2016).

Wind vs Seismic Comparison: Base shear from wind = 98.56 kN; from earthquake = 95.76 kN. Both are comparable; however, wind governs for external cladding/wall design while seismic governs inertial loading of the structural frame. Both must be checked independently per code.

10. Summary of Design Forces

Member	Load Type	Factored Load / Force	Governing Combo
Slab	Dead + Live	$w_u = 9.56 \text{ kN/m}^2$; $M_u = 29.9 \text{ kN}\cdot\text{m/m}$	1.5(DL+LL)
Beam (5 m)	Dead + Live	$w_u = 45.69 \text{ kN/m}$; $V_u = 114.2 \text{ kN}$; $M_u = 142.8 \text{ kN}\cdot\text{m}$	1.5(DL+LL)
Column (GF)	Dead + Live	$P_u = 1,603.5 \text{ kN}$	1.5(DL+LL)
Wind Base Shear	Wind	$V_{\text{base}} = 98.56 \text{ kN}$	1.5(DL+WL)
Seismic Base Shear	Earthquake	$V_B = 95.76 \text{ kN}$; $A_h = 0.048$	1.2(DL+LL+EQ)

Conclusion: All structural elements are within safe design limits under IS code provisions. The gravity load combination 1.5(DL+LL) governs beam and column design. Both wind and seismic lateral loads produce comparable base shears ($\approx 96\text{--}99 \text{ kN}$) and must both be checked. Seismic detailing per IS 13920 is mandatory

for Zone III. Further detailed design should include flexural and shear reinforcement calculation, footing design, and inter-storey drift verification.

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